

# DOTUN OLAOYE

Columbia, Missouri | Permanent Resident (Green Card Holder)  
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## SUMMARY

PhD student in Statistical Genetics specializing in quantitative genetics and genomic prediction across animal and plant breeding systems. Experienced in genome-phenome association analysis, trait prediction, and mixed-model genetic evaluation using both frequentist and Bayesian statistical frameworks. Background in plant pathology and disease-resistance breeding, with three years of industry experience evaluating crop disease resistance within a commercial trait-advancement pipeline.

## EDUCATION

### **Doctor of Philosophy - Statistical Genetics** (with Dr. Jared Decker)

University of Missouri, Columbia. June 2024 - Present

Research focus: multi-breed genomic prediction of puberty and fertility traits in beef heifer cattle

### **Master of Science - Plant Breeding, Genetics and Pathology**

University of Arkansas, Fayetteville. May 2021

### **Bachelor of Agriculture** (Honor's: Crop Protection/Pathology)

Federal University of Agriculture, Abeokuta, Nigeria. 2017

## WORK EXPERIENCE

**Graduate Research Assistant**, University of Missouri, Columbia. June 2024 - Present

- Built and validated a multi-breed genomic prediction pipeline (univariate, bivariate, and multi-trait single-step GBLUP and GBLUP) for puberty and fertility traits across five beef cattle breed populations (Angus, Red Angus, Hereford, Bos indicus-influenced, and a pooled MultiBreed population of over 9,000 animals genotyped on an 800K SNP panel) using BLUPF90+/GIBBSF90+
- Estimated variance components, heritability, and genetic correlations via REML/AI-REML, and validated genomic prediction accuracy using a Linear Regression (LR) temporal validation framework
- Presented genomic prediction research at the Advances in Genome Biology and Technology (AGBT) Agricultural Meeting, 2025, and the Population, Evolutionary, and Quantitative Genetics Conference (PEQG), 2026

**Collaborative Research Project** (ongoing, independent of PhD assistantship duties, originated during MS), University of Arkansas collaborators

- Extended weighted genomic prediction methodology to disease-resistance and abiotic-stress trait prediction in cowpea and common bean, using SNP weighting informed by meta-transcriptomic (RNA-seq) evidence

**Pathology Research Specialist**, Pathology Unit, Trait Assessment, North America, Syngenta Seeds Inc. May 2021 - April 2024

- Assessed field corn, sweet corn, and soybean for resistance or tolerance to plant pathogens of economic importance, feeding disease trait data directly into product advancement decisions
- Planned and implemented pathology research trials on field corn within the Trait Assessment program
- Served as a member of Market Segment Teams for Field Corn pathology traits in North America
- Handled the adoption of a proprietary data collection tool for field corn pathology trials

**Program Technician**, Vegetable Breeding and Genetic Laboratory, Department of Horticulture, University of Arkansas, Fayetteville. April 2021 - May 2021

- Worked on improving economically important traits in spinach, arugula, soybean, and common bean

**Graduate Research Assistant**, spinach breeding and genetics for downy mildew resistance, University of Arkansas, Fayetteville. January 2019 - April 2021

- Bred spinach for resistance to the downy mildew pathogen *Peronospora effusa*, including genetic characterization of resistance loci and race-specific resistance across worldwide germplasm
- Conducted disease assays and phenotyping for disease and agronomic traits in field corn, sweet corn, soybean, and snap bean, in field and greenhouse settings
- Produced high-quality inoculum for disease assays and standardized disease assay protocols

**Laboratory Attendant**, Plant Pathology Unit, International Institute of Tropical Agriculture, IITA-CGIAR, Ibadan, Nigeria. November 2018 - January 2019

- Biocontrol management strategies for black Sigatoka disease of banana plants

**Graduate Research Unpaid Internship**, Plant Pathology Unit, International Institute of Tropical Agriculture, IITA-CGIAR, Ibadan, Nigeria. June 2018 - November 2018

- Characterized the biology of plant pathogens causing diseases of economic importance in field corn and banana

**Undergraduate Research Assistantship**, Department of Crop Protection, Federal University of Agriculture, Abeokuta, Nigeria. June 2015 - October 2016

- Participated in molecular marker evaluation of indigenous tomato genotypes for Fusarium wilt resistance (CAPS marker TAO1, endonuclease FOK1, homozygous I2 resistance gene selection)

## **LEADERSHIP**

Genetics Students' Representative, Graduate Professional Council, University of Missouri-Columbia

Trained new hires and supervised seasonal technical personnel and a research associate within the Field Corn Pathology team, Trait Assessment, North America, Syngenta Seeds Inc.

Currently mentoring an honors student in the research lab

Mentored a bachelor's student on manuscript preparation and publication (see acknowledgment: <https://doi.org/10.1080/03235408.2020.1715756>)

## **PROFESSIONAL SERVICE**

Served as Ad-hoc Secretary for the Vigna spp USDA's CGC at the Annual International Crop Science Meeting, Texas, USA. November 2019

Judge, Agronomy and Horticulture 4-H program, University of Minnesota. July 2022

Invited as a panelist for the 2023 Professional Development Workshop, Corteva Symposium, West Africa

Volunteer Reviewer: Agrosystems, Geosciences & Environment; Nigerian Journal of Biotechnology

- Provided reviews on three manuscripts for the journal, Nigerian Journal of Biotechnology
- Provided reviews on a few manuscripts for the journal, Agrosystems, Geosciences & Environment

Reviewed at least 10 conference abstracts for Plant Health 2024, American Phytopathological Society

Member and volunteer, African Student Organization, University of Arkansas, Fayetteville. 2019-2021

Volunteer member, Volunteer Action Center, Fayetteville, Arkansas. 2019-2021

## KEY SKILLS

**Quantitative genetics/genomic prediction:** Single-step GBLUP (ssGBLUP) and GBLUP via BLUPF90+/GIBBSF90+, including threshold models for binary traits; univariate, bi-variate, and multi-trait mixed-model analyses; both frequentist (REML/AI-REML) and Bayesian (Gibbs sampling/MCMC via GIBBSF90+) variance component estimation; Linear Regression (LR) validation; genomic relationship matrix construction (VanRaden method); weighted/evidence-informed genomic prediction

**Population genetics:** Principal component analysis and population stratification correction (PLINK); genotype imputation (Beagle); variant filtering and manipulation (bcftools)

**Programming:** R, Python, Perl, SQL, SAS, JMP, Quarto/R Markdown, shell scripting, HPC job orchestration (SLURM)

**Genomic/bioinformatics tools:** BLUPF90+/GIBBSF90+ family, PLINK, bcftools, Beagle, TASSEL, MEGA, QTL mapping and GWAS software

**Laboratory and field:** Inoculum production for disease assays, disease assay standardization, phenotypic data collection, marker-trait association analysis, tissue collection for genomic and phenotypic analysis

## AWARDS

**FFAR Fellow** (Stipend + Professional Development; ~\$148,500 over 3 years: student stipend, tuition, and fees), Foundation for Food and Agriculture Research, in partnership with North Carolina State University. 2025-2028 (see: [FFAR Fellows story](#))

- Highly competitive fellowship designated for high-potential scientists, funding 150+ hours of professional development and leadership training
- Training activities included Professional Development Plans, Mentoring, Project Management, Designing and Managing Meetings, Facilitating Participatory Decision-Making, Conflict Management, Cross-Cultural Dynamics, 360-degree Leadership, Effective Mentoring, Crucial Conversations and Crucial Influencing, Productivity Workshop, Effective Science Communication, and Policy Communication, and a visit to the United States Congress to advocate for agricultural research funding

**2025 Scholars of Distinction**, University of Missouri Graduate School (recognizing external funding recipients). 2025

**PhD Graduate Research Assistantship**, Genetics Area Program, University of Missouri, Columbia. 2024 - Present

**MS Graduate Research Assistantship** (~\$66,830), Department of Horticulture, University of Arkansas, Fayetteville. 2019-2021 (Funded by a USDA grant to work on a Specialty Crop)

**2020 Conference Travel Grant Award** (\$350), American Society for Horticultural Science

**Two-time (2019 & 2020) Travel Grant Award** (\$600), Graduate School, University of Arkansas, Fayetteville

**2019 Plant Health and Quality Summer School Scholarship** (€1200), Campus Vegetal of Angers, University of Angers, France

## PUBLICATIONS

**Olaoye, D.**, Bhattarai, G., Correll, J., Shi, A. Genetic analysis reveals novel genomic regions for downy mildew resistance and validates the potential for predictive breeding in spinach (*Spinacia oleracea*). Under review, BMC Plant Biology, 2026.

**Olaoye, D.**, Rasaki, L., Adesina, O., Kareem, B., Kandel, S., Ravelombola, W., Yang, Y., Shi, A. RNA-seq meta-analysis and machine learning identify stress-responsive genes and improve genomic prediction in common bean (*Phaseolus vulgaris* L.) with cross-species application in cowpea (*Vigna unguiculata* L.). bioRxiv (2026). <https://doi.org/10.64898/2026.08.08.743654> (preprint)

**Olaoye, D.**, Bhattarai, G., Feng, C. et al. Evaluation of downy mildew resistance in spinach (*Spinacia oleracea*). Euphytica 220, 38 (2024). <https://doi.org/10.1007/s10681-023-03289-9>

Bhattarai G, **Olaoye D**, Mou B, Correll JC and Shi A (2022) Mapping and selection of downy mildew resistance in spinach cv. whale by low coverage whole genome sequencing. Front. Plant Sci. 13:1012923. doi: 10.3389/fpls.2022.1012923

Zia, B, A. Shi, **D. Olaoye**, H. Xiong, W. Ravelombola, P. Gepts, H.F. Schwartz, M.A. Brick, K. Otto, B. Ogg, and S. Chen. 2022. Genome-wide association study and genomic prediction for bacterial wilt resistance in common bean (*Phaseolus vulgaris*) core collection. Front. Genet. <https://doi.org/10.3389/fgene.2022.853114>

Ravelombola, W., Shi, A., Huynh, B.L., Qin, J., Xiong, H., Manley, A., Dong, L., **Olaoye, D.**, Bhattarai, G., Zia, B. and Alshaya, H., 2022. Genetic architecture of salt tolerance in a Multi-Parent Advanced Generation Inter-Cross (MAGIC) cowpea population. BMC genomics, 23(1), pp.1-22.

Ravelombola, W., Dong, L., Barickman, T.C., Xiong, H., **Olaoye, D.**, Bhattarai, G., Zia, B., Alshaya, H., Alatawi, I. and Shi, A., 2021. Evaluation of salt tolerance in cowpea at seedling stage. Euphytica, 217(6), pp.1-20.

Ravelombola, W., Shi, A., Chen, S., Xiong, H., Yang, Y., Cui, Q., **Olaoye, D.** and Mou, B., (2020). Evaluation of cowpea for drought tolerance at seedling stage. Euphytica, 216(8), pp.1-19.

Yang, Y., Shi, D., Wang, Y., Zhang, L., Chen, X., Yang, X. ... & Yang, G. (2020). Transcript profiling for regulation of sweet potato skin color in Sushu8 and its mutant Zhengshu20. *Plant Physiology and Biochemistry*, 148, 1-9.

## PRESENTATIONS

*Presenter (poster):* Olaoye, D., Franklin, G., Schnabel, R., Thomas, J., Decker, J. Predicting Puberty and Fertility Traits in Multi-breed Beef Heifer Cattle. Population, Evolutionary, and Quantitative Genetics Conference (PEQG), 2026

*Co-author (poster):* Franklin, G., Olaoye, D., Thomas, J., Decker, J. Genome-wide association studies of puberty and fertility traits in US beef heifers. Population, Evolutionary, and Quantitative Genetics Conference (PEQG), 2026

*Presenter:* Estimating genetic parameters and predicting puberty and fertility traits in multi-breed beef heifer cattle. Advances in Genome Biology and Technology (AGBT) Agricultural Meeting, 03/31-04/2, 2025

*Presenter:* Olaoye, D., Bhattarai, G., Feng, C., Correll, J. C., & Shi, A. (2020, August). Evaluation of Resistance to Pfs race 5 in Worldwide Spinach Collection. In 2020 ASHS Annual Conference. ASHS.

*Presenter:* Olaoye, D., Shi, A., Ravelombola, W. S., Bhattarai, G., & Xiong, H. (2020, August). Drought Tolerance Characterization in two Contrasting Common Bean Genotypes. In 2020 ASHS Annual Conference. ASHS.

*Presenter:* Olaoye, D., Bhattarai, G., Feng, C., Shi, A., & Correll, J. (2019, November). Evaluation of Incomplete Dominance in a Spinach Downy Mildew Resistance Locus. In ASA, CSSA and SSSA International Annual Meetings (2019).

*Co-author:* Yang, Y., Cui, Q., Qin, J., Ravelombola, W., Bhattarai, G., Zia, B., ... & Shi, A. (2019, November). Evaluation of Drought Tolerance in Common Bean at Seedling Stage. In ASA, CSSA and SSSA International Annual Meetings (2019).

*Co-author:* Effect of Exogenous Application of Plant Growth Regulator in the Control of *Fusarium oxysporum* f. sp. *lycopersici* using two Tomato Genotypes. International Conference of the Biotechnology Society of Nigeria, 2018

*Co-author:* CAPS Marker TAO1 and Endonuclease FOK1 in the Selection of Tomato Genotypes with Homozygous I2 Resistance Gene to Fusarium Wilt. International Conference of the Biotechnology Society of Nigeria, 2018

## PROFESSIONAL MEMBERSHIP AND TRAINING

Current Graduate Student Member: Genetics Society of America (GSA)

Former Graduate Student Member: American Society of Horticultural Science; American Society of Agronomy; Crop Science Society of America; Soil Science Society of America

Bruce Weir Summer Institute in Statistical Genetics (SISG). Summer 2025

Modules completed: QG1 (Quantitative Genetics), QG2 (Mixed Models for Genetic Analysis), HE3 (Artificial Intelligence/Machine Learning)

5th Annual Meeting of the Arkansas Bioinformatics Consortium (AR-BIC 2019). Bioinformatics in Food and Agriculture, UAMS, Little Rock, AR. February 25-26, 2019

Molecular biology workshop, Inqaba Biotec West Africa, University of Ibadan. April 2018

## REFERENCES

### **Zachary King, PhD**

Corn North Regional Pipeline Manager and Corn Breeder, Syngenta

Collaborator, Syngenta Seeds Inc.

Contact information available upon request

### **Amy Baetsen-Young, PhD**

Trait Development Lead, Syngenta

Collaborator, Syngenta Seeds Inc.

Contact information available upon request